

02__baseline__ocr

March 24, 2026

1 02 Baseline OCR

This notebook is where I check that the core OCR path is doing something sensible before I worry about cleanup or reporting. The goal is not to make the page look pretty; it is to see whether the crop and line split are at least pointing the model at the right text.

```
[ ]: import sys
      from pathlib import Path

      cwd = Path.cwd().resolve()
      ROOT = cwd if (cwd / 'src').exists() else cwd.parent
      if str(ROOT) not in sys.path:
          sys.path.insert(0, str(ROOT))

      from PIL import Image
      import matplotlib.pyplot as plt

      from src.layout import crop_array, extract_main_text_region, segment_lines
      from src.ocr_model import TrOCRPrintedBaseline
      from src.preprocessing import prepare_line_for_ocr, preprocess_page
```

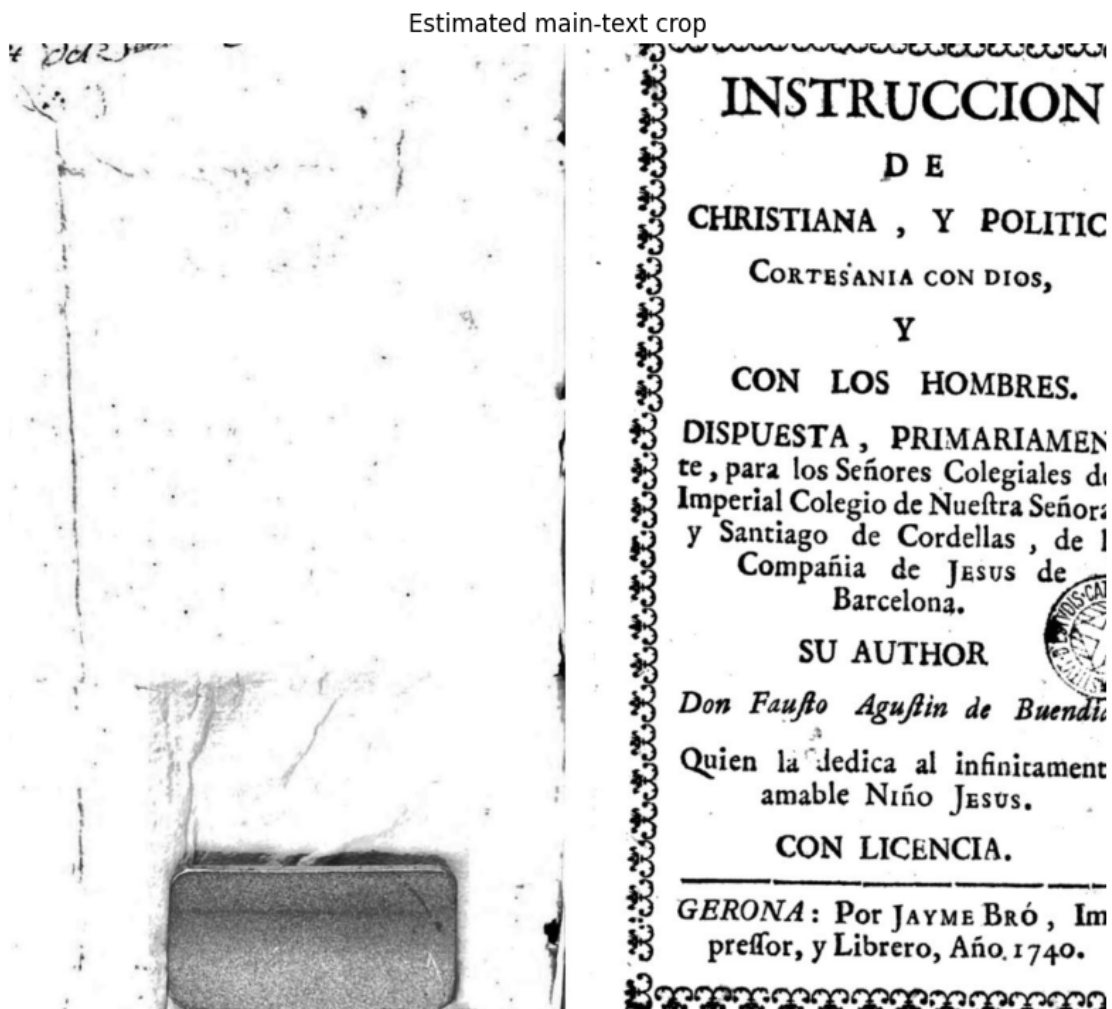
```
[ ]: page_paths = sorted((ROOT / 'data' / 'page_images').glob('*.png'))
      page_path = page_paths[0]
      image = Image.open(page_path).convert('RGB')
      views = preprocess_page(image)
      crop = extract_main_text_region(views['gray'], views['binary'])
      gray_crop = crop_array(views['gray'], crop)
      binary_crop = crop_array(views['binary'], crop)
      line_arrays = segment_lines(gray_crop, binary_crop)
      page_path.name, crop.reason, len(line_arrays)
```

```
[ ]: ('buendia_instruccion_page_0001.png', 'fallback_small_crop', 26)
```

A quick visual check usually tells me more than the first metric run. If the crop is already drifting into side notes here, the OCR result will not mean much.

```
[ ]: plt.figure(figsize=(10, 12))
      plt.imshow(gray_crop, cmap='gray')
```

```
plt.axis('off')
plt.title('Estimated main-text crop');
```



```
[ ]: model = TrOCRPrintedBaseline()
line_images = [prepare_line_for_ocr(line) for line in line_arrays[:10]]
predictions = model.predict_lines(line_images)
predictions[:5]
```

Loading weights: 0% | 0/478 [00:00<?, ?it/s]

VisionEncoderDecoderModel LOAD REPORT from: microsoft/trocr-base-printed

Key	Status
encoder.pooler.dense.bias	MISSING
encoder.pooler.dense.weight	MISSING

Notes:

- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

```
[ ]: ['4 HRX X 1.1%',  
      '@0.74 3 INSTRUCTION',  
      'WWW.WWWW.W.W.M.',  
      'UNIMUM P 1 V W',  
      'LIAT TAKELL & CHRISTANA, Y POLITIC']
```